

# YOUSSEF EL HOUSNI

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New Jersey, USA

yelhousni.github.io

## EDUCATION

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| PhD at <b>Ecole Polytechnique, Paris</b><br>Cryptography (Computer Science and Mathematics)<br>Team GRACE<br>Advisors: François Morain, Daniel Augot, and Aurore Guillevic<br>Dissertation: <i>The arithmetic of pairing-based proof systems</i> | November 2019 – December 2022 |
| MEng at <b>ENSEIRB-MATMECA, Bordeaux</b><br>Electronics and Electrical Engineering<br>Majors: Image and Signal Processing                                                                                                                        | 2013 – 2016                   |
| Preparatory Classes (MPSI/MP) at <b>Moulay Hassan CPGE, Tangier</b><br>Mathematics and Physics                                                                                                                                                   | 2011 – 2013                   |

## WORK EXPERIENCE

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| <b>Consensys</b><br><i>Staff Cryptography Researcher</i> <ul style="list-style-type: none"><li>· Applied cryptography research with a focus on zero-knowledge proofs.</li><li>· Development of high-performance cryptographic libraries in Go.</li><li>· Core contributor to gnark and gnark-crypto.</li><li>· Working on Linea, a type-2 zkEVM.</li></ul>                      | September 2023 – Present<br><i>New Jersey, USA</i>  |
| <b>Consensys</b><br><i>Staff Cryptography Researcher</i> <ul style="list-style-type: none"><li>· Applied cryptography research with a focus on zero-knowledge proofs.</li><li>· Development of high-performance cryptographic libraries in Go.</li><li>· Core contributor to gnark and gnark-crypto.</li><li>· Contributed to the cryptographic foundations of Linea.</li></ul> | December 2020 – August 2023<br><i>Paris, France</i> |
| <b>EY Blockchain Hub</b><br><i>Senior Cryptography Engineer</i> <ul style="list-style-type: none"><li>· Applied cryptography research on zero-knowledge proofs and proxy re-encryption.</li><li>· Blockchain development with a focus on privacy (Rust, C++, Solidity).</li><li>· Contributed to Baseline and Nightfall.</li></ul>                                              | August 2018 – November 2020<br><i>Paris</i>         |
| <b>Secure-IC</b><br><i>R&amp;D Engineer</i> <ul style="list-style-type: none"><li>· Applied cryptography research on elliptic curves, pairings, and SCA/FIA countermeasures.</li><li>· Video steganography on embedded devices (up to UHD@60fps 10-bit).</li><li>· Conception of stochastic models and statistical tests for hardware TRNGs and PUFs.</li></ul>                 | October 2016 – August 2018<br><i>Rennes</i>         |
| <b>Orange Labs</b><br><i>Research Intern</i>                                                                                                                                                                                                                                                                                                                                    | March 2016 – August 2016<br><i>Bordeaux, France</i> |

- Research internship on applied cryptography and signal processing topics.
- Worked on engineering research problems in secure communications and data processing.
- Produced prototypes and technical analyses in support of the research team.

**Kansai University**

*Research Intern*

June 2015 – August 2015

*Osaka, Japan*

- Research internship in an academic lab environment.
- Worked on image and signal processing problems in collaboration with the host research group.
- Implemented and evaluated experimental methods for the internship project.

## TECHNICAL AND SOFT SKILLS

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|------------------|------------------------------------------------------------------------------------|
| <b>Software</b>  | Go, C/C++, Python, Rust, Solidity, SageMath, Bash, L <sup>A</sup> T <sub>E</sub> X |
| <b>Languages</b> | English (fluent), French (bilingual), Arabic (bilingual)                           |

## PUBLICATIONS

- [1] **Youssef El Housni** and Benedikt Bünz, *On the Security of Constraint-Friendly Map-to-Curve Relations*, ePrint 2026/590. [ia.cr/2026/590](#)
- [2] **Youssef El Housni**, *Fast cube roots in  $Fp2$  via the algebraic torus*, ePrint 2026/392. [ia.cr/2026/392](#)
- [3] Dimitri Koshelev, **Youssef El Housni**, and Georgios Fotiadis, *Batch subgroup membership testing on pairing-friendly curves*, ePrint 2025/1311. [ia.cr/2025/1311](#)
- [4] Liam Eagen, **Youssef El Housni**, Simon Masson, and Thomas Piellard, *Fast elliptic curve scalar multiplications in  $SN(T)ARK$  circuits*, LATINCRYPT 2025. ePrint 2025/933. DOI: [10.1007/978-3-032-06754-8\\_4](#)
- [5] Gautam Botrel and **Youssef El Housni**, *Faster Montgomery multiplication and Multi-Scalar Multiplication for SNARKs*, IACR Transactions on Cryptographic Hardware and Embedded Systems, 2023(3). ePrint 2022/1400. DOI: [10.46586/tches.v2023.i3.504-521](#)
- [6] Antonio Sanso and **Youssef El Housni**, *Families of prime-order endomorphism-equipped embedded curves on pairing-friendly curves*, Journal of Cryptology, 2024. ePrint 2023/1662. DOI: [10.1007/s00145-024-09514-5](#)
- [7] **Youssef El Housni**, *Pairings in Rank-1 Constraint System*, ACNS 2023. ePrint 2022/1162. DOI: [10.1007/978-3-031-33488-7\\_13](#)
- [8] Diego F. Aranha, **Youssef El Housni**, and Aurore Guillevic, *A survey of elliptic curves for proof systems*, Designs, Codes and Cryptography, 2022. ePrint 2022/586. DOI: [10.1007/s10623-022-01135-y](#)
- [9] **Youssef El Housni**, Aurore Guillevic, and Thomas Piellard, *Co-factor clearing and subgroup membership testing on pairing-friendly curves*, AFRICACRYPT 2022. ePrint 2022/352. DOI: [10.1007/978-3-031-17433-9\\_22](#)
- [10] **Youssef El Housni** and Aurore Guillevic, *Families of SNARK-friendly 2-chains of elliptic curves*, EUROCRYPT 2022. ePrint 2021/1359. DOI: [10.1007/978-3-031-07085-3\\_13](#)
- [11] **Youssef El Housni**, *The arithmetic of pairing-based proof systems*, PhD dissertation, Ecole Polytechnique, 2022. [hal.science/tel-03922488v2](#)
- [12] **Youssef El Housni** and Aurore Guillevic, *Optimized and secure pairing-friendly elliptic curves suitable for one layer proof composition*, CANS 2020. ePrint 2020/351. [ia.cr/2020/351](#)

- [13] J.-L. Danger, **Y. El Housni**, A. Facon, C.T. Gueye, S. Guilley, S. Herbel, O. Ndiaye, E. Persichetti, and A. Schaub, *On the Performance and Security of Multiplication in  $GF(2^N)$* , Cryptography, MDPI, Volume 2, 2018. DOI: 10.3390/cryptography2030025
- [14] Sylvain Guilley and **Youssef El Housni**, *Random Numbers Generation: Tests and Attacks*, FDTC 2018, pp. 49–54. DOI: 10.1109/FDTC.2018.00016

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### PATENTS

- [9] **Youssef El Housni** and Florent Lozac’h, *Device and method for testing a sequence generated by a random number generator*, WO2020136142A1

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### SELECTED TALKS

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|----------|-----------------------------------------|--------------------------------------------------------------|
| Jan 2026 | Morocco Blockchain Days, Rabat, Morocco | <i>ZK Proofs and Blockchain Scalability</i>                  |
| Oct 2025 | LATINCRYPT 2025, Medellín, Colombia     | <i>Fast EC Scalar Multiplications in SN(T)ARKs</i>           |
| Jul 2025 | Google Seminar, NYC, USA                | <i>On pairing implementation</i>                             |
| Apr 2025 | UM6P Seminar, Ben Guerir, Morocco       | <i>Zero-knowledge proofs and blockchains</i>                 |
| Nov 2024 | Devcon 7, Bangkok, Thailand             | <i>Elliptic curves and SNARKs — Past, present and future</i> |

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### AWARDS

**1st Place**, ZPrize 2022 – Mobile category (zero-knowledge proof competitions)

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### EXTRA-CURRICULAR

- Co-organizer of “ZK-PAR” – a technical meetup in Paris on privacy-preserving technology in blockchains.
- Played in the Moroccan soccer regional championship with Tetouan FC (2000–2011).